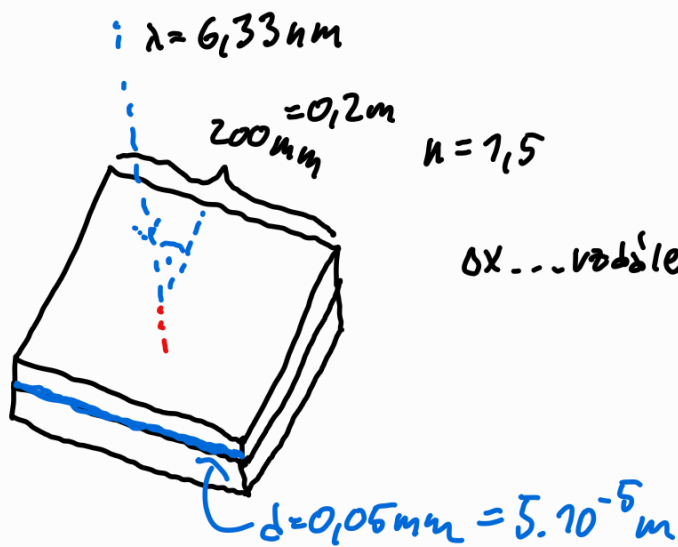
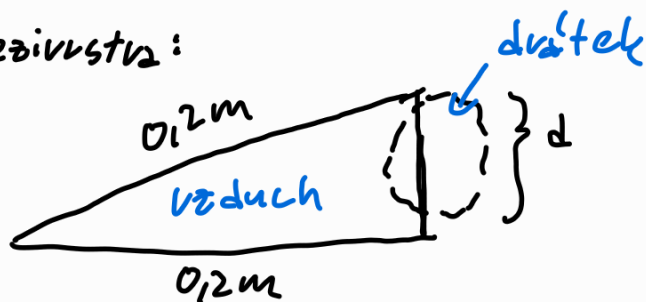


79)



$\Delta x \dots$ vzdálenost sv. proměněn interf. obrazce

mezivrstva:



$$\tan \alpha = \frac{d}{z} = \frac{5 \cdot 10^{-5}}{0.2} = 2.5 \cdot 10^{-4}$$

$$t(x) = \frac{d}{z} x$$

podm. interference:

$$\Delta L = \left(k + \frac{1}{2}\right) \lambda ; k = 1, 2, \dots$$

$$\Delta L = z + t(x) = z + \frac{d}{z} x$$

$$2 \frac{d}{z} x = \left(k + \frac{1}{2}\right) \lambda$$

$$x_k = \frac{\left(k + \frac{1}{2}\right) \lambda}{2} \cdot \frac{z}{d}$$

$$\Delta x = x_{k+1} - x_k$$

$$\Delta x = \left(\frac{\left(k+1 + \frac{1}{2}\right) \lambda}{2} - \frac{\left(k + \frac{1}{2}\right) \lambda}{2} \right) \frac{z}{d}$$

$\frac{\lambda}{2}$

$$\Delta x = \frac{\lambda}{2} \frac{z}{d} = \frac{633 \cdot 10^{-9} \cdot 0.2}{2 \cdot 5 \cdot 10^{-5}} = \underline{\underline{1.266 \cdot 10^{-3} \text{m}}}$$